

Predictive Modeling and A/B Testing Report

IBM Telco Customer Churn Dataset

Introduction

Customer churn, the rate at which customers stop doing business with a company, is one of the most consequential problems in subscription-based industries. For telecom providers, acquiring a new customer typically costs far more than retaining an existing one, making the ability to identify at-risk customers before they leave enormously valuable. This project applies machine learning to the IBM Telco Customer Churn dataset with two goals: first, to build and evaluate classifiers that predict which customers are likely to churn; and second, to go a step further and test whether those predictions translate into a practical, measurable advantage in a simulated retention campaign.

The central research question guiding this project is: does targeting high-risk customers using machine learning reduce churn more effectively than a random retention strategy? To answer it, three classification models are trained and compared. Logistic regression, random forest, and a support vector machine with an RBF kernel, and the best-performing model's predictions are used to design a simulated A/B experiment.

Dataset

The dataset used in this project is the IBM Telco Customer Churn dataset, publicly available on Kaggle. It contains records for 7,043 customers of a fictional telecommunications company, each described by 21 features spanning four broad categories: demographics (gender, senior citizen status, partner and dependent status), account information (tenure, contract type, billing method, payment method), services subscribed (phone, internet, streaming, online security, tech support, and others), and financials (monthly charges and total charges). The target variable is a binary indicator of whether the customer churned.

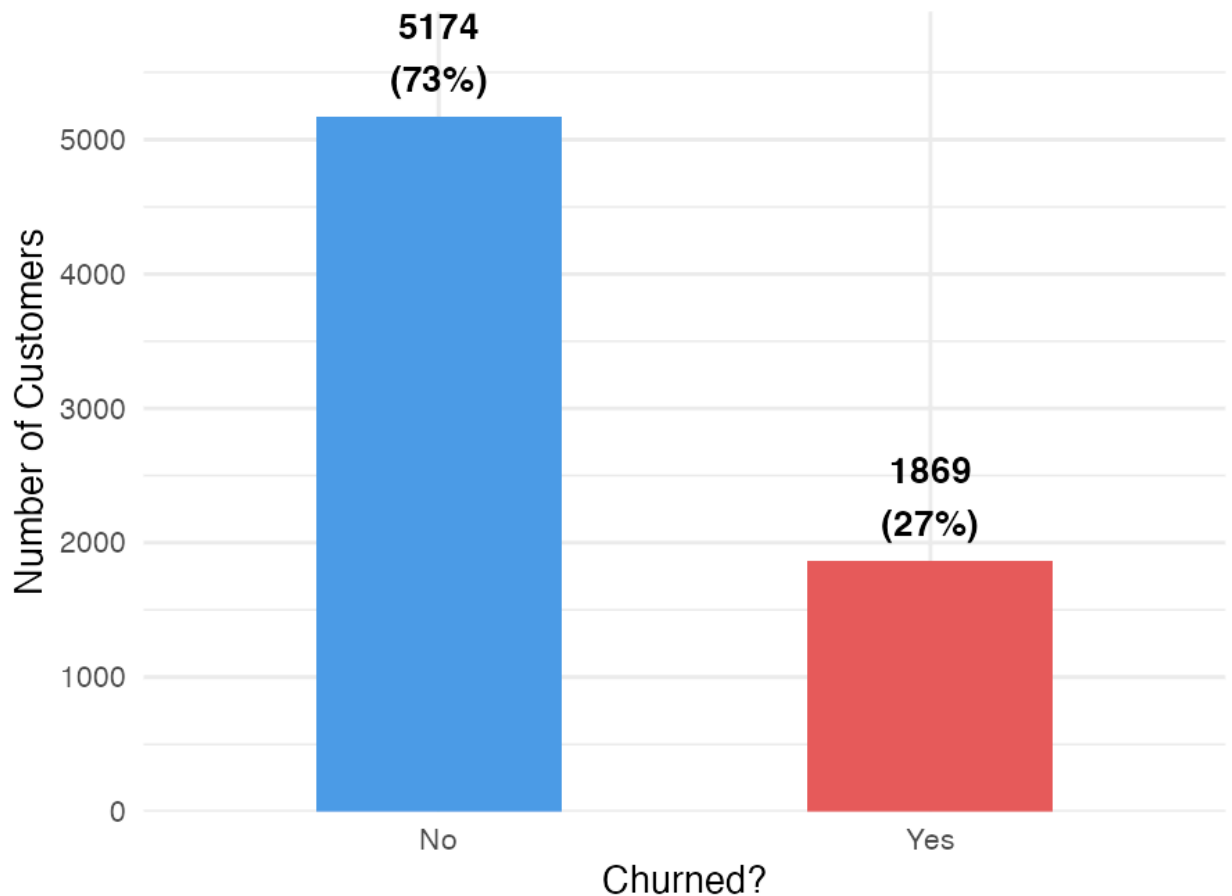
Of the 7,043 customers, 1,869 churned, an overall churn rate of approximately 26.5%. This class imbalance is an important consideration throughout the analysis. A naive model that predicts no churn for every customer would achieve 73% accuracy without any predictive value, which is why accuracy alone is insufficient as an evaluation metric and why AUC-ROC, precision, recall, and F1 score are used alongside it.

Exploratory Data Analysis

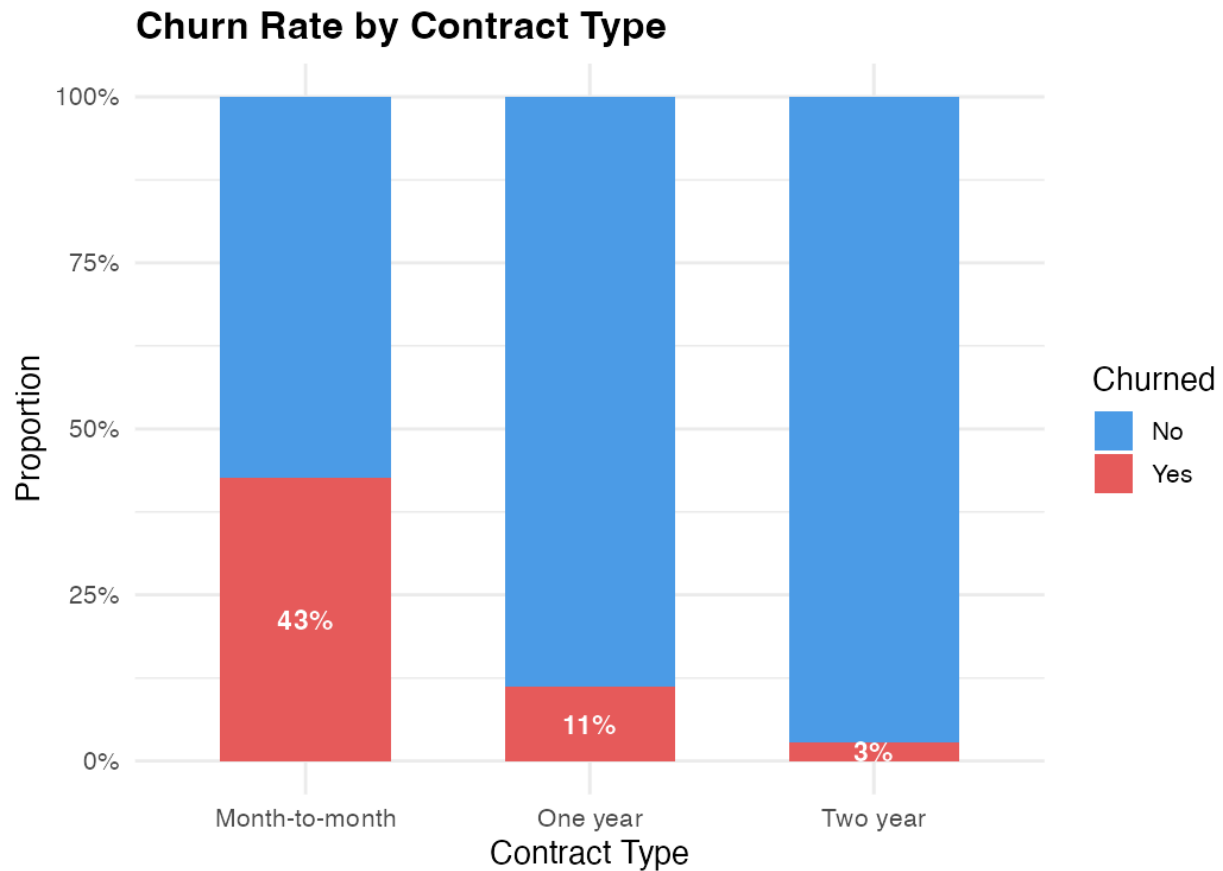
Before modeling, the data was explored to understand the distribution of churn across key customer segments. Several strong patterns emerged that informed both the preprocessing strategy and the interpretation of model results.

Customer Churn Distribution

Overall churn rate: 26.5%



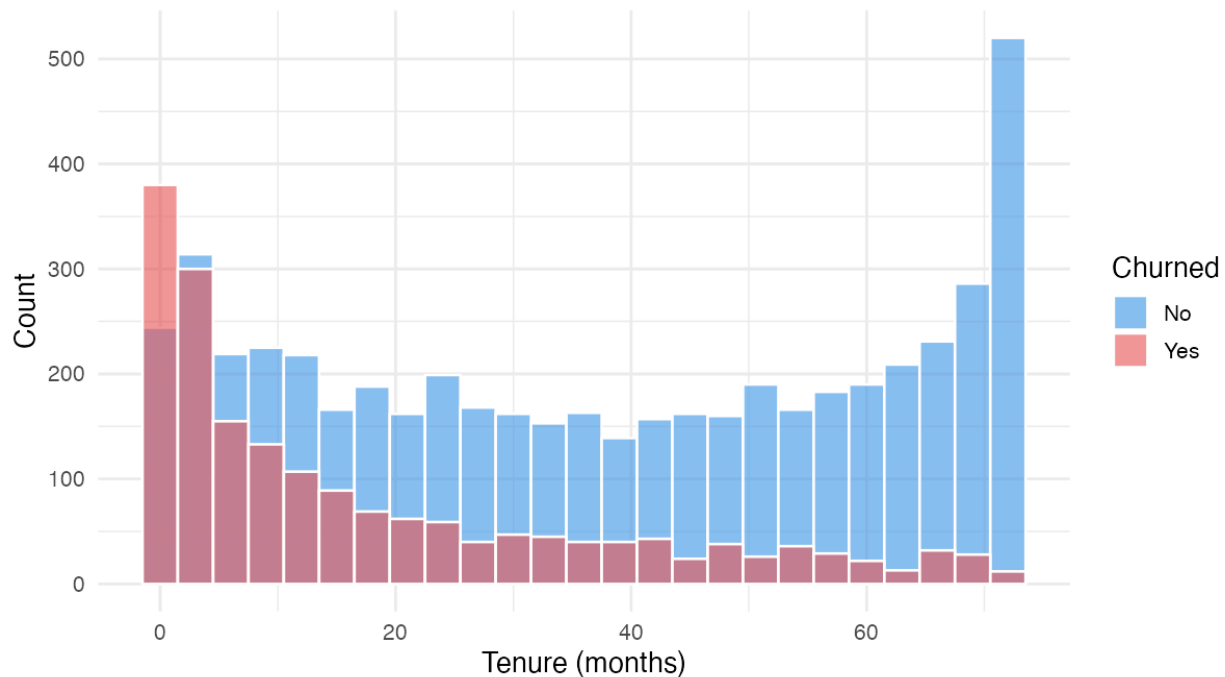
Contract type proved to be the single strongest predictor of churn. Customers on month-to-month contracts churned at a rate of 43%, compared to just 11% for one-year contracts and 3% for two-year contracts. This pattern suggests that customers who have not committed to a long-term agreement are far more likely to leave, possibly because they face no financial penalty for doing so and have greater flexibility to switch providers.



Tenure, the number of months a customer has been with the company, shows a clear inverse relationship with churn. Customers who churned were heavily concentrated in the early months of their tenure, while long-tenured customers rarely left. This is consistent with the idea that customers who stay past a certain threshold become increasingly loyal, either due to inertia, switching costs, or genuine satisfaction with the service.

Tenure Distribution by Churn Status

Churned customers tend to leave early



Monthly charges also showed a meaningful association with churn. Customers who churned paid higher monthly bills on average, suggesting that price sensitivity plays a role in the decision to leave. Payment method was another informative variable, customers using electronic check churned at 45%, nearly three times the rate of those on automatic payment methods (15-17%). Customers paying automatically may simply be less likely to actively reconsider their subscription. Among service features, customers subscribed to tech support and online security had notably lower churn rates, while streaming service subscribers showed rates close to the overall average.

Preprocessing

Several preprocessing steps were applied to prepare the data for modeling. The customer ID column was removed as it carries no predictive information. A small number of blank entries in the TotalCharges column, belonging to new customers with zero tenure, were imputed to zero. The SeniorCitizen variable, originally stored as a 0/1 integer, was left in numeric form consistent with the other binary features.

Binary categorical variables such as gender, partner status, and paperless billing were encoded as 0 and 1. Multi-level categorical variables including contract type, internet service type, and payment method, were one-hot encoded using R's `model.matrix()` function, which creates a separate binary column for each category.

Column names produced by this process were cleaned using the janitor package to remove spaces and special characters.

The data was split into a training set (80%) and a test set (20%) using stratified sampling, ensuring that both sets reflect the original churn rate of approximately 26.5%. Finally, the three numeric features, tenure, monthly charges, and total charges, were standardized using z-score normalization. Critically, the scaling parameters (mean and standard deviation) were computed on the training set only and then applied to both sets. This prevents data leakage, ensuring that the test set remains a true simulation of unseen data and that evaluation metrics are not optimistically biased.

Methods

Three classification models were trained: logistic regression, random forest, and a support vector machine with a radial basis function (RBF) kernel. Logistic regression serves as an interpretable linear baseline, it estimates the log-odds of churn as a weighted combination of the input features and is straightforward to explain. Random forest is an ensemble method that builds 500 decision trees, each trained on a random subset of the data, and aggregates their predictions. This approach handles non-linear relationships and feature interactions naturally. The SVM with RBF kernel finds the maximum-margin boundary between churners and non-churners in a transformed feature space, making it effective when the classes are not linearly separable in the original space.

All models were trained using 5-fold cross-validation, in which the training data is divided into five equal partitions. In each round, four partitions are used to train the model and the fifth is held out for validation. This process repeats five times so that every data point is used for validation exactly once, and performance is averaged across all five rounds. The metric optimized throughout was AUC-ROC, chosen because it evaluates model performance across all possible classification thresholds and is robust to class imbalance.

Hyperparameter tuning was conducted within the cross-validation loop. For random forest, the number of features considered at each split (*mtry*) was tuned across values of 3, 5, 7, and 10. For the SVM, the regularization parameter *C* (tested at 0.1, 1, and 10) and the kernel width sigma (tested at 0.01, 0.05, and 0.1) were tuned jointly. Logistic regression has no tuning parameters in this configuration. All modeling was implemented in R using the caret package.

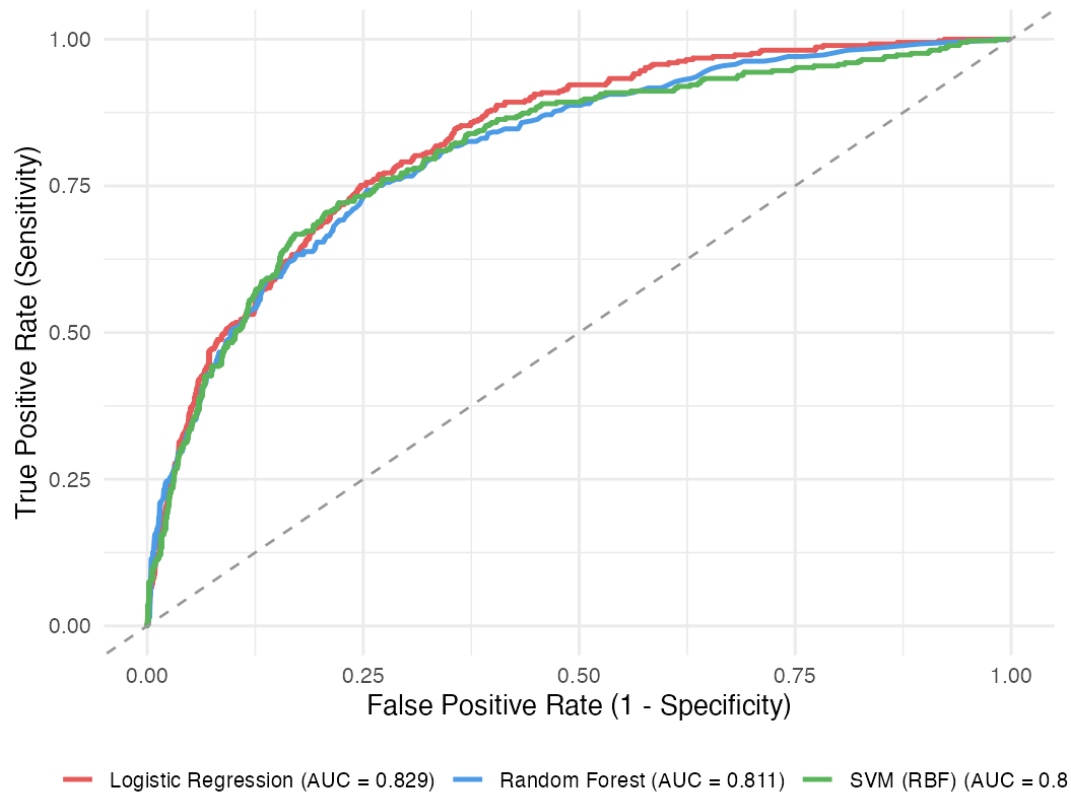
Results

Model	AUC	Accuracy	Precision	F1	Recall
Logistic Regression	0.829	0.799	0.331	0.571	0.499
Random Forest	0.811	0.799	0.337	0.527	0.429
SVM (RBF)	0.810	0.799	0.327	0.534	0.442

All three models achieved comparable performance, with AUC scores ranging from 0.810 to 0.829. Logistic regression led with an AUC of 0.829, slightly outperforming both random forest (0.811) and the SVM (0.810). This result is notable, the simpler and more interpretable model performed as well as or better than the more complex ensemble and kernel-based approaches on this dataset. Accuracy was identical across all three models at 79.9%, which reflects the class imbalance, all models tend toward predicting the majority class. Recall for the churn class ranged from 43% to 50%, indicating that roughly half of true churners are missed. This is a known limitation of models trained on imbalanced data without threshold adjustment.

ROC Curves -- Test Set

All models vs. random classifier (dashed)



A/B Testing

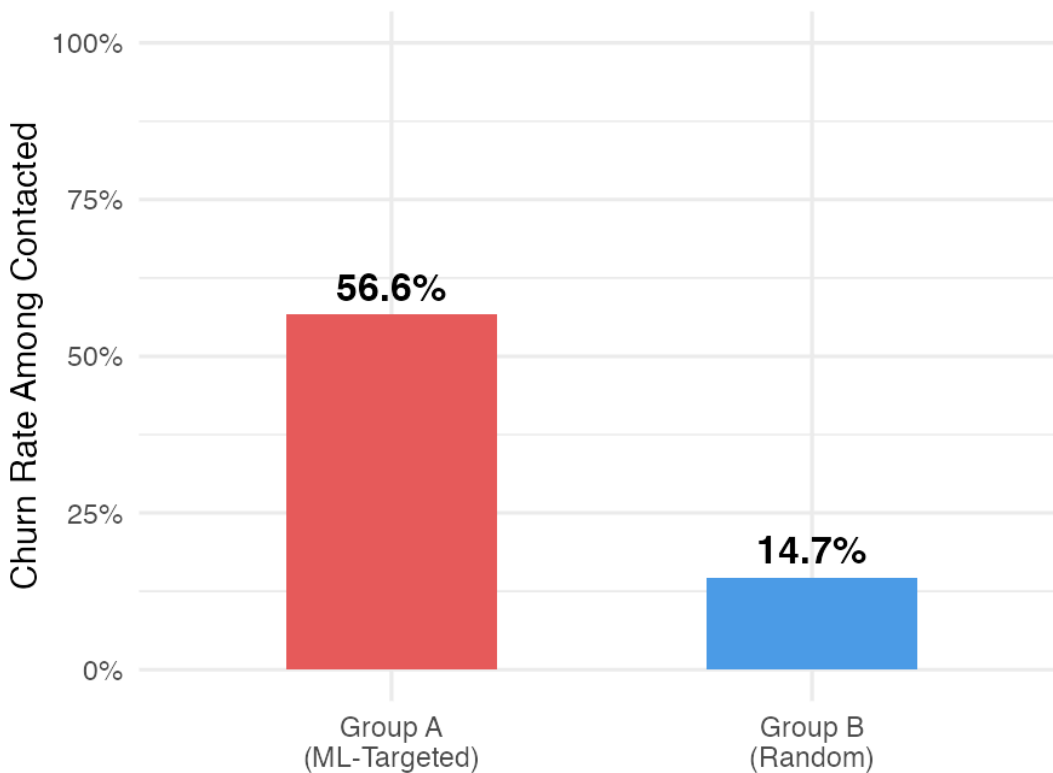
After identifying the best-performing model, a simulated A/B experiment was conducted to evaluate whether ML-driven targeting produces better outcomes than random customer selection in a retention campaign. The test set served as the simulated customer pool. A campaign budget constraint of 30% was imposed, representing a scenario where the company can afford to contact approximately 422 customers with a retention offer.

Customers were divided into two groups of equal size. Group A (ML-Targeted) consisted of the top 30% of customers ranked by predicted churn probability from the logistic regression model. Group B (Random) consisted of 422 customers selected uniformly at random from the remaining pool. The outcome measured was the churn rate within each group, that is, of the customers contacted, what proportion actually churned.

The results were clear. Group A had a churn rate of 56.6%, compared to 14.7% in Group B. This means the ML-targeted group identified nearly four times as many true churners per customer contacted. The absolute difference of approximately 42 percentage points was statistically significant ($p < 0.001$) by both a two-proportion z-test and a chi-square test of independence. A bootstrap confidence interval (2,000 resamples) on the absolute risk difference confirmed this result was not due to chance. The lift curve showed that within the 30% campaign window, the model achieved approximately 2.1 times the churn identification rate of random targeting. These results support the conclusion that ML-based targeting provides a meaningful practical advantage over undirected outreach.

Churn Rate: ML-Targeted vs. Random

Campaign reach = 30% of customers | n = 422 per group



Conclusion

This project demonstrated that machine learning can effectively identify customers at risk of churning from a telecommunications provider, and that these predictions translate into a measurable advantage in a targeted retention scenario. Exploratory analysis revealed that contract type, tenure, and monthly charges are the strongest drivers of churn, with month-to-month customers churning at 43% compared to 3% for those on two-year contracts.

Among the three models evaluated, logistic regression achieved the highest AUC of 0.829 on the held-out test set, slightly outperforming random forest and SVM despite its simplicity. This outcome highlights an important lesson: more complex models do not always outperform simpler ones, especially on structured tabular data with a moderate number of features. The random forest model's feature importance rankings confirmed that tenure and total charges are the most predictive variables, reinforcing the EDA findings.

The A/B testing simulation showed that directing retention efforts toward the highest-risk customers resulted in finding 3.9 times more true churners per customer

contacted compared to random outreach. At a 30% campaign budget, the model achieved a lift of approximately 2.1x over the random baseline, a result that was statistically significant across multiple tests.

A key limitation of this analysis is that the A/B test is simulated rather than conducted in a real environment. In practice, the effectiveness of a retention campaign also depends on the offer made, the channel used, and whether customers actually respond. Additionally, the models' recall for churners was relatively low (43-50%), meaning a substantial portion of at-risk customers were not flagged. Future work could explore threshold adjustment, oversampling techniques such as SMOTE, or cost-sensitive learning to improve sensitivity to the minority class.